

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: ARABICMMLU | **Context:** Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Label-correctness verification process — 96% inter-annotator agreement on 100 sampled questions [Q42, Q43] with native MSA speakers holding Bachelor's degrees and competitive compensation [Q35, Q36, Q39] — establishes a reasonable practical ceiling [Q44]. However, annotator-pool composition undermines convergent validity for the multi-country deployment: 10 annotators are from Jordan (4), Egypt (2), Lebanon (1), UAE (1), KSA (2), with no Moroccan, Kuwaiti, or Palestinian annotators [Q31] despite Palestine being a top-3 source country [WEB-3, WEB-4]. For contested historical questions where 'correct' answers vary across national curricula, the 96% agreement likely reflects Jordanian/Egyptian consensus rather than pan-regional validity. The sampled data shows concrete examples of Jordanian-curriculum framing for shared Arab history [DATASET-D1, DATASET-D2, DATASET-D3, DATASET-D11]. Training data contamination is acknowledged as unverifiable [Q95]. OCR corruption in university Egyptian items [DATASET-D23, DATASET-D24] suggests label quality may degrade in some sub-corpora. No aggregation method considerations for minority perspectives are documented.

ID	Evidence
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q35	"Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)
Q36	"We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)
Q39	"While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q95	"to a lack of sufficient information about its training regimen. As such, we cannot assert that the model's pretraining data is free from contamination." (p.10)
D1	دحو مامت: ني طسلف يف اهعاطأو ةينوي هصلال ةكرحل نم لوأل هلل ادبع كل مل فق او م نم تسيل ةيتأل نم ةدحاو هضف / ي ني طسلف ةمأل قودنص معد يل برعل ةودع / قارب طئاح يل ارمم دوهي اعطع ا هضف / م 1950 اع ني تفصلال Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective
D2	نب لص يف ري مأل ٢ / ني سحل نب هلل ادبع ري مأل ١: سي راب يف ءاف ل حل هديق يذل حل صل رم توم يف برعل _____ ل ثم Jordan civics HS: who represented Arabs at Paris Peace Conference — Jordanian Hashemite framing
D3	Jordan civics HS: بادتال كف دوي ق نم دال بال ررح: ١٩٢٨ ماع هي ناطي ربال هي ندرأل هدهاع ملل سيئرل فدهل ناك primary goal of the 1948 Jordanian-British treaty — Jordanian national framing
D11	Jordan primary social science: fact about King Talal's reign — Jordan-specific في كل ذو ةي حص باب سأل هتيالو تهت ن ني سحل نب لال ط كل مل
D23	Severely garbled OCR text making question almost illegible ة زكرم لا: يل ري تدت ... صا خال أل ني تب ق طل تدل ا عتي هت ةت جرد

D24	قـبـس ام لك؟ مل اـرـسـالـا رـلـوـدـل اـرـف مـكـار لـا رـف اـر رـفـاـوت بـرر رـتـل اـرـسـالـا طـوـرـرـشـل اـنـرـم — Egypt university political science with severe OCR corruption — content partially unreadable
WEB-3	Palestine confirmed top-3 source country — yet no Palestinian annotators in pool (https://github.com/mbzuai-nlp/ArabicMMLU)
WEB-4	HuggingFace dataset card confirms Palestine major contributor (https://huggingface.co/datasets/MBZUAI/ArabicMMLU)

Strengths

- 96% answer-key accuracy verification on a 100-question sample provides a documented quality ceiling [Q42, Q43, Q44]
- Native-speaker annotator workforce with at least Bachelor's degrees and competitive pay [Q35, Q36, Q39]
- Pre-collection workshop standardized annotator process [Q40]
- Annotators include workers from 5 of 8 deployment countries (Jordan, Egypt, Lebanon, UAE, KSA) [Q31, Q96]

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Q35	"Our internal workers are Master's students and Research Assistants in Computer Science, while the external workers hold Bachelor's degrees." (p.4)
Q36	"We ensured competitive compensation for the workers, exceeding the monthly average wage in each respective country." (p.4)
Q39	"While our workers are native speakers of Modern Standard Arabic with at least Bachelor's degrees, we maintain the quality of our dataset construction through meticulous steps." (p.4)
Q40	"Firstly, we conducted a 1-hour workshop before the data collection stage to clarify the process." (p.4)
Q42	"Finally, we assessed the accuracy of our data collection by having two native Arabic speakers annotate 100 randomly sampled questions. They were provided with all metadata, including the answer key, and tasked with verifying the correctness of each sample using any available resources (e.g., search engines)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)
Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Partially — labels reflect curriculum-stated 'correct' answers from source exams, which encode national perspectives. For non-contested STEM content this is acceptable [DATASET-D21, DATASET-D38]; for contested civic/historical content [DATASET-D1, DATASET-D3], labels reflect source-country curriculum perspective without acknowledgment.

OC-2: High potential for disagreement on contested content — Palestinian, Moroccan, and Kuwaiti perspectives are not represented in the verifier pool [Q31] despite Palestine being a major source [WEB-3, WEB-4]. The 96% figure [Q43] likely overstates regional validity for politically sensitive items.

OC-3: Documented: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 UAE, 2 KSA), 4 external workers (3 Jordanian, 1 Egyptian) [Q31]; internal workers are Master's/RA-level CS students, external workers hold Bachelor's [Q35]. Demographics are documented but skew Jordanian/Egyptian.

OC-4: Re-annotation by Palestinian, Moroccan, and Kuwaiti annotators recommended for contested-content items, especially history/civics from those source countries.

OC-5: No aggregation method for minority/competing perspectives is documented — the benchmark's structure precludes it [Q27].

OC-6: Documented: (1) annotator-pool skew toward Jordan/Egypt with absent Moroccan/Palestinian/Kuwaiti perspectives [Q31, Q96]; (2) 96% ceiling does not account for cross-curriculum disagreement on contested content [Q43, Q44]; (3) OCR corruption in some sub-corpora [DATASET-D23, DATASET-D24]; (4) contamination unverifiable [Q95].

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
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Q96	"We extend our gratitude to all collaborators from Jordan, Egypt, Lebanon, UAE, and Saudi Arabia who participated in the data collection process." (p.10)
D1	دحو و مامت: ني طس ل ف ي ف اعم اطأو ة ني نو ي ه ص ل ا ة ك ر ح ل ا ن م ل و أ ل ا ه ل ل ا د ب ع ك ل م ل ا ف ق ا و م ن م ت س ي ل ة ي ت أ ل ا ن م ة د ح و ه ص ل ف ر / ي ني ط س ل ف ل ا ة م أ ل ا ق و د ن ص م غ د ي ل ا ب ر ع ل ا ة و ع د / ق ا ر ب ل ا ط ئ ا ح ي ل ا ا ر م د و ه ي ل ا ا ط ع ل ا ه ص ل ف ر / م 1950 م ا ع ن ي ت ف ص ل ا ا — Jordan HS history question about King Abdullah I's positions on Zionism and Palestine — framed from Jordanian curriculum perspective
D3	Jordan civics HS: ب ا د ت ن ا ل ا ك ف د و ي ق ن م د ا ل ب ل ا ر ح ت : ١٩٢٨ م ا ع ه ي ن ا ط ي ر ب ل ا ه ي ن د ر أ ل ا ه د ا ع م ل ل س ي ئ ر ل ا ف د ه ل ا ن ا ك primary goal of the 1948 Jordanian-British treaty — Jordanian national framing
D21	ت ا د ي ب ي ل ل ا ع ي ن ص ت ن ع ة ل و ئ س م ة ي ل ا ت ا ل ت ا ي ض ع ل ا ي د ح / X100 : و ه ي ئ و ض ل ا ر ه ج م ل ا ي ف ة ي ت ي ز ل ا ة س د ع ل ا ر ي ب ك ت ر ا د ق م — Palestine HS biology: standard cellular biology — culturally neutral STEM
D23	Severely garbled OCR text making question almost illegible — ... ص ا خ ت ل أ ل ا ن ي ت ب ع ط ل ت د ل ا ع ت ي ي ه ت ة ت ج ر د
D24	Egypt university political science with severe OCR corruption — content partially unreadable — ق ب س ا م ل ك م ا ر ر س ا ل ا ر ل و د ل ا ر ف م ك ا ر ر ل ا ر ف ا ر ر ر ف ا و ت ب ر ر ر ر ت ل ا ر ر س ا ل ا ط و ر ر ر ش ل ا ن ر ر م
D38	+ م و ي س ي ن غ ا م ل ا ل ص ح ي ذ ل ا ل ع ا ف ت ل ا ف ص ت ة ي ل ا ت ا ل ت ا ل د ا ع م ل ا ي أ ؛ ر ب ت خ م ل ا ب م و ي س ي ن غ ا م ل ا ن م ة ي م ك ق ر ح ب د م ا م ا ق — Natural science MS: chemistry reactions — country-neutral STEM
WEB-3	Palestine confirmed top-3 source country — yet no Palestinian annotators in pool (https://github.com/mbzuai-nlp/ArabicMMLU)
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Information Gaps

- Whether the 100-question verification sample included contested-content items
- Distribution of OCR corruption across the full dataset

Requires Expert Verification

- Whether Palestinian, Moroccan, and Kuwaiti subject-matter experts would agree with ground-truth labels on country-sourced contested items

Remediation

Gap 1: Annotator pool lacks Palestinian, Moroccan, and Kuwaiti members despite Palestine being a top-3 source country [Q31, WEB-3]

Recommendation: Re-verify a stratified sample of Palestinian-, Moroccan-, and Egyptian-sourced contested-content items with annotators from those countries; report disagreement rates separately from the 96% pooled figure to expose curriculum-specific disagreement.

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